Lecture 6.1

Image Restoration

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Image Degradation Model

- It represents how an original image gets degraded due to noise, blur, or other distortions during acquisition or transmission.
- The degraded image $g(x,y)$ is modeled as:
- $g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$
- Where:
- $f(x,y)$ = original image
- $h(x,y)$ = degradation function (e.g., blur kernel)
- $*$ = convolution operator
- $\eta(x,y)$ = additive noise (e.g., Gaussian noise)
- $g(x,y)$ = observed degraded image

Model of Image Degradation/Restoration Process

The degraded image in frequency domain for 2-D Fourier transform G(u,v) is modeled as:

$$G(u,v) = H(u,v) * F(u,v) + N(u,v)$$

Spatial domain

$$g(x,y) = (h \star f)(x,y) + \eta(x,y)$$

Frequency domain

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

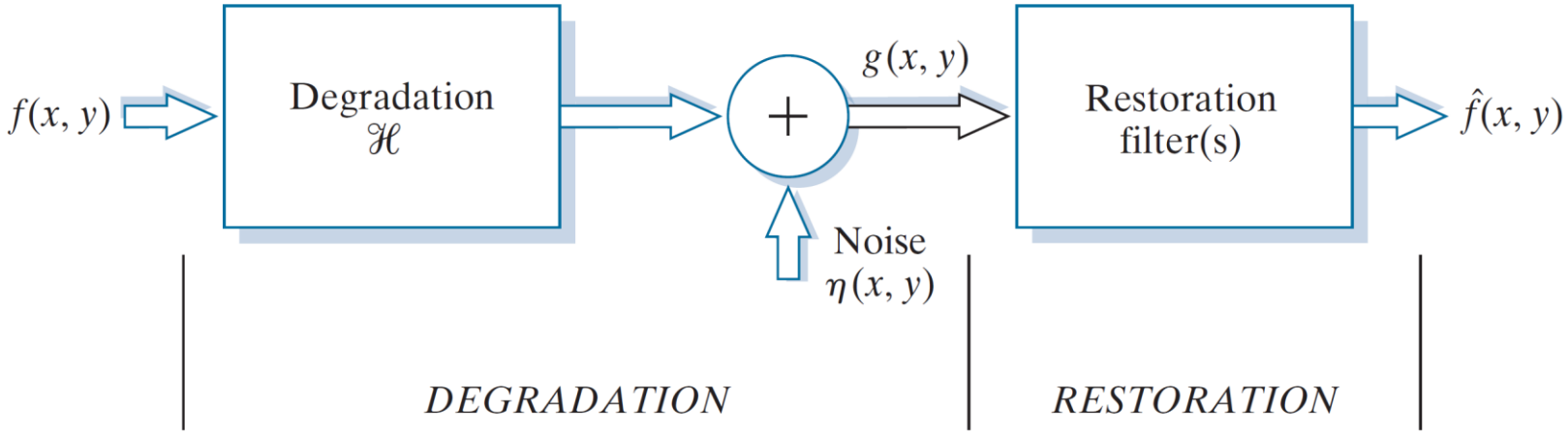


Image Degradation Model

Step 1: Original Image $f(x, y)$

- Let's take a simple 1D "image" vector for clarity (can imagine a 1-row image):

$$f = [10,20,30,40,50]$$

Step 2: Degradation Function $h(x, y)$

- Suppose degradation is a blurring effect modeled by a simple averaging kernel of size 3:

$$h = \frac{1}{3}[1, 1, 1]$$

- This means each pixel in degraded image is average of itself and neighbors.

Image Degradation Model

Step 3: Convolution $h * f$: Calculate blurred version:

- Position 1: average of $f1, f2$ (assuming zero padding before $f1$):

$$(0 + 10 + 20)/3 = 30/3 = 10$$

Position 2: average of $f1, f2, f3$:

$$(10 + 20 + 30)/3 = 60/3 = 20$$

Position 3: average of $f2, f3, f4$:

$$(20 + 30 + 40)/3 = 90/3 = 30$$

Position 4: average of $f3, f4, f5$:

$$(30 + 40 + 50)/3 = 120/3 = 40$$

Position 5: average of $f4, f5$ (zero padding after $f5$):

$$(40 + 50 + 0)/3 = 90/3 = 30$$

Thus, $h * f = [10,20,30,40,30]$

Image Degradation Model

Step 4: Add Noise $\eta(x,y)$

Suppose noise vector is: $\eta=[1,-2,1,0,-1]$

Step 5: Degraded Image g

Add noise to blurred image:

$g=[10+1,20-2,30+1,40+0,30-1]=[11,18,31,40,29]$

Summary:

Pixel Position	Original f	Blur $h * f$	Noise η	Degraded g
1	10	10	+1	11
2	20	20	-2	18
3	30	30	+1	31
4	40	40	0	40
5	50	30	-1	29

Restoration in Presence of Noise OnlySpatial Filtering

$$g(x, y) = f(x, y) + \eta(x, y)$$

$$G(u, v) = F(u, v) + N(u, v)$$

Corrupted Image

Restored Image

Noise Image

Noise Model

- ✓ The **Noise Model** is the companion concept to the **Image Degradation Model**.
- ✓ In image processing, **noise** means **unwanted random variations of brightness or color** in an image.
- ✓ It corrupts the true signal, making the image look grainy, spotty, or distorted.

Additive Noise Model $g(x,y) = f(x,y) + \eta(x,y)$

$$G(u,v) = F(u,v) + N(u,v)$$

Multiplicative Noise Model $g(x,y) = f(x,y).\eta(x,y)$

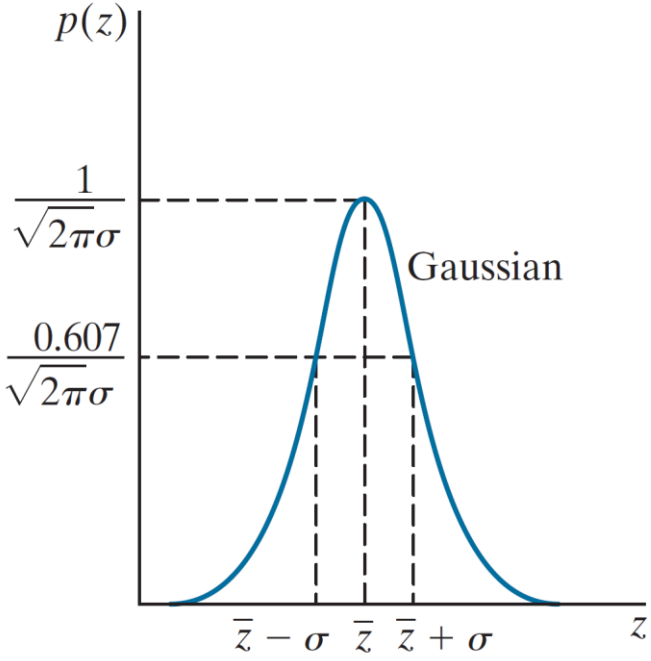
Noise Model

1. Gaussian Noise
2. Uniform Noise
3. Exponential Noise
4. Rayleigh Noise
5. Erlang (Gamma) Noise
6. Poisson (Photon) Noise:
7. Speckle Noise:
8. Salt-and-Pepper Noise

1. Gaussian Noise

pdf:

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\bar{z})^2}{2\sigma^2}} \qquad -\infty < z < \infty$$



~75%: $\bar{z} - \sigma, \bar{z} + \sigma$

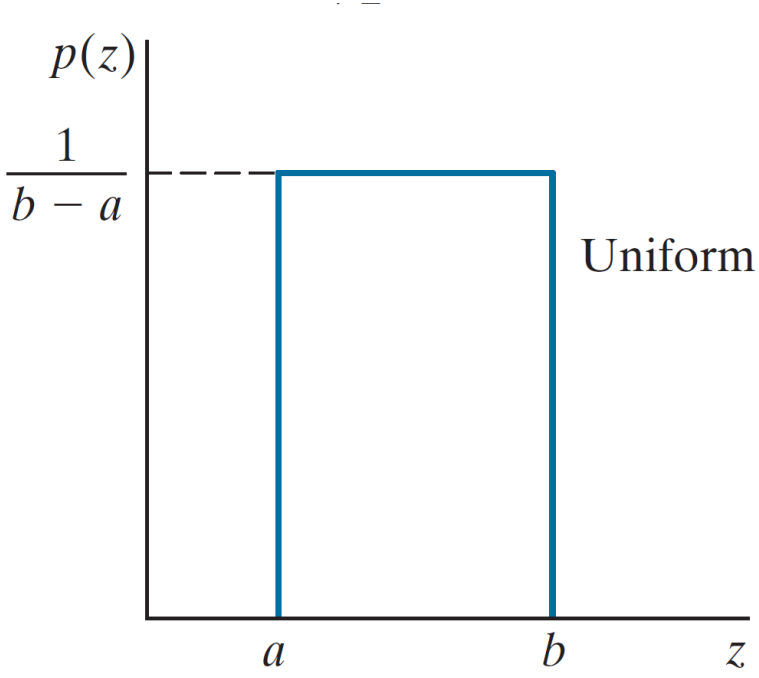
~95%: $\bar{z} - 2\sigma, \bar{z} + 2\sigma$

2. Uniform Noise

pdf:

$$p(z) = \begin{cases} \dfrac{1}{b-a} & a \le z \le b \\ 0 & \text{otherwise} \end{cases}$$

$$\bar{z} = \frac{a+b}{2} \qquad \sigma^2 = \frac{(b-a)^2}{12}$$



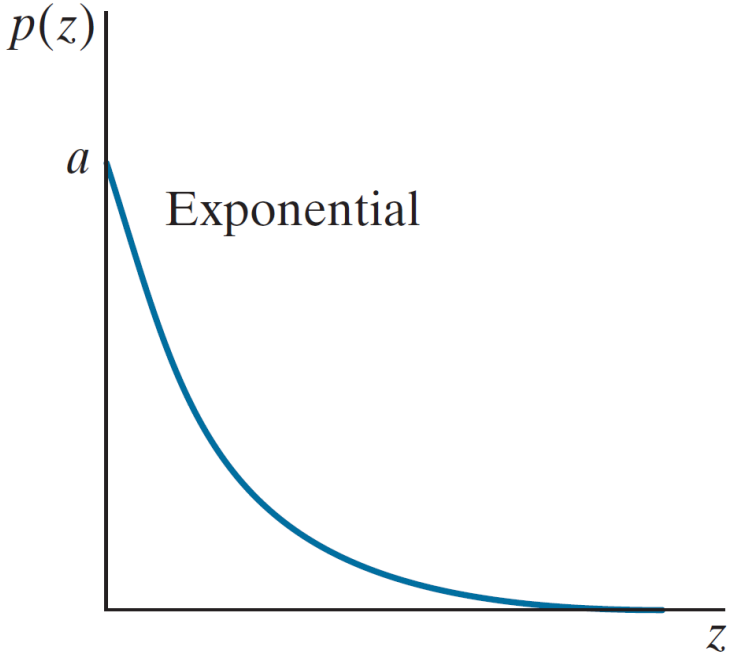
3. Exponential Noise

pdf:

$$p(z) = \begin{cases} ae^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

$a > 0, b = 1$

$$\bar{z} = \frac{1}{a} \qquad \sigma^2 = \frac{1}{a^2}$$



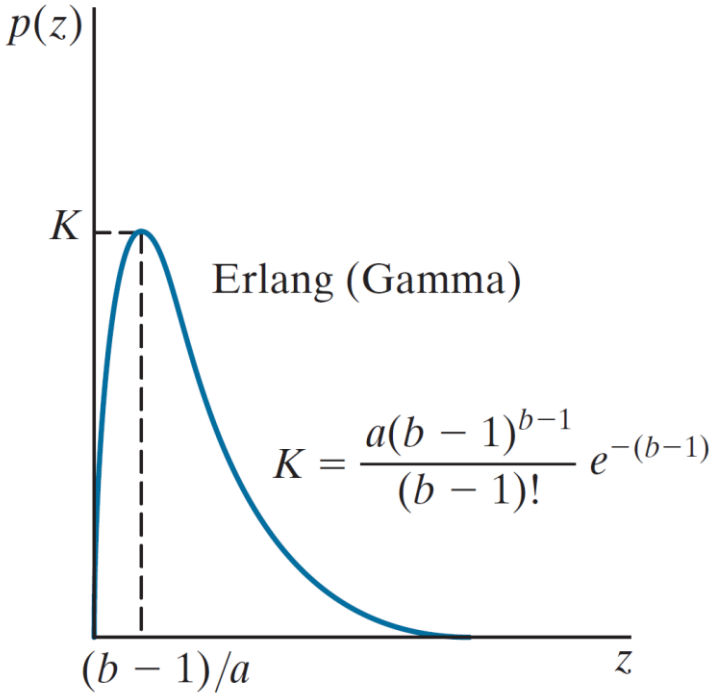
4. Erlang (Gamma) Noise

pdf:

$$p(z) = \begin{cases} \dfrac{a^b z^{b-1}}{(b-1)!} e^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

$a, b > 0$

$$\bar{z} = \frac{b}{a} \qquad \sigma^2 = \frac{b}{a^2}$$



5. Rayleigh Noise

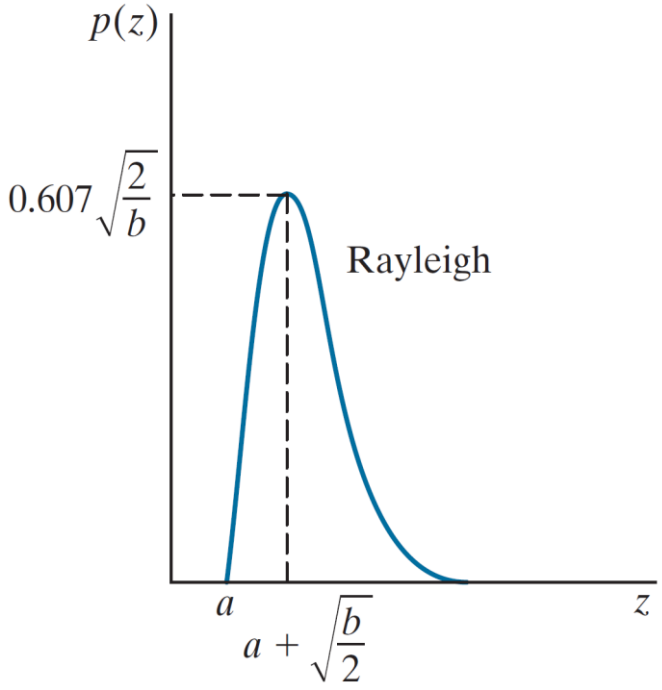
pdf:

$$p(z) = \begin{cases} \frac{2}{b}(z - a)e^{-(z - a)^2/b} & z \geq a \\ 0 & z < a \end{cases}$$

$$\bar{z} = a + \sqrt{\pi b/4}$$

$$\sigma^2 = \frac{b(4 - \pi)}{4}$$

Skewed to right



6. Poisson (Photon) Noise

$$p(z) = \frac{\lambda^z e^{-\lambda}}{z!}$$

Example: Occurs in photon-limited imaging (e.g., X-ray, CT)

vii) Speckle Noise:

$$p(z) = \frac{1}{\sigma^2} z e^{-z^2/2\sigma^2}$$

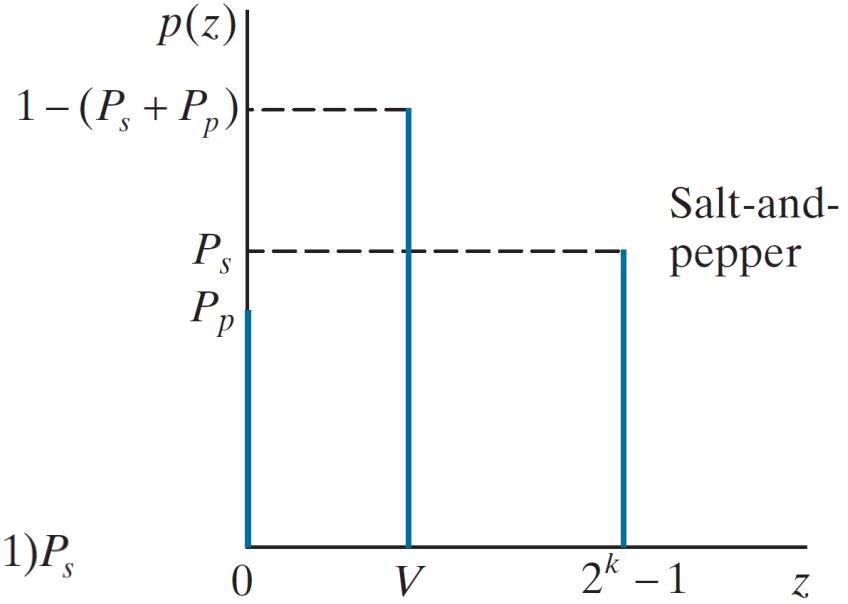
Example: Multiplicative noise (SAR (Synthetic Aperture Radar), ultrasound)

8. Salt-and-Pepper (Impulse) Noise

pdf (bipolar) impulse noise:

$$p(z) = \begin{cases} P_s & \text{for } z = 2^k - 1 \\ P_p & \text{for } z = 0 \\ 1 - (P_s + P_p) & \text{for } z = V \end{cases}$$

$$0 < V < 2^k - 1$$



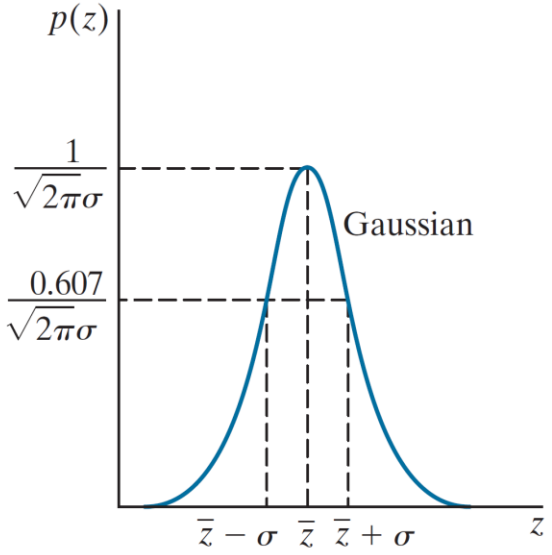
The mean of salt-and-pepper noise is given by

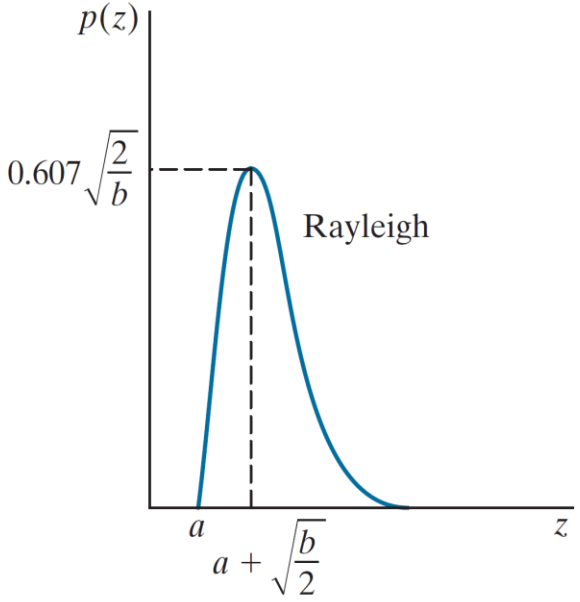
$$\bar{z} = (0)P_p + K(1 - P_s - P_p) + (2^k - 1)P_s$$

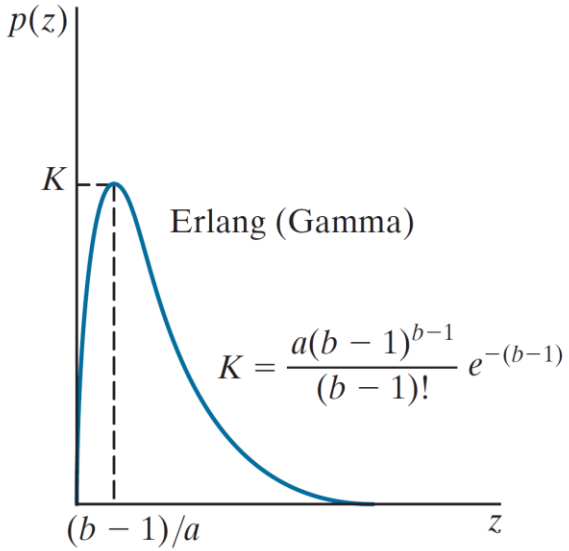
$$\sigma^2 = (0 - \bar{z})^2 P_p + (K - \bar{z})^2 (1 - P_s - P_p) + (2^k - 1)^2 P_s$$

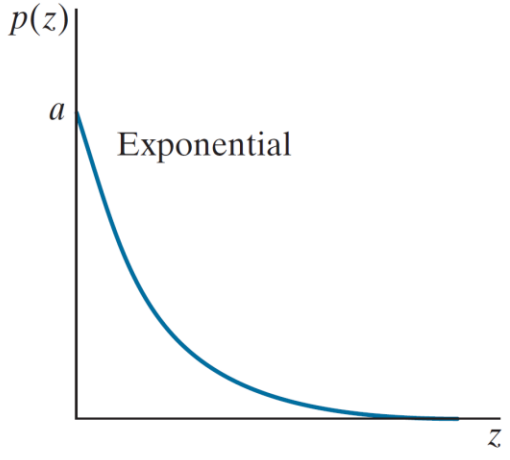
Example: Random black and white pixels; due to bit errors or dead pixels.

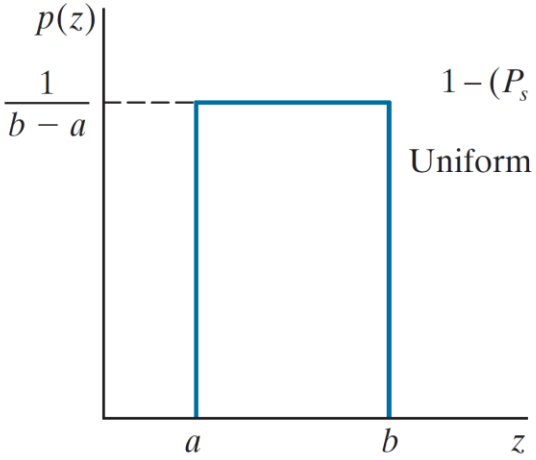
Noise Model

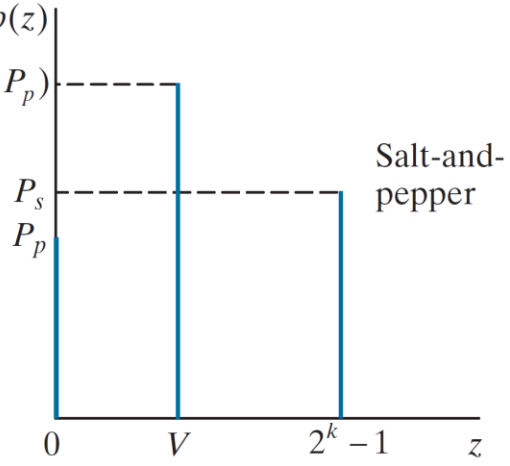

p(z)
1/√(2πσ)
0.607/√(2πσ)
Gaussian
z̄ − σ
z̄
z̄ + σ
z


p(z)
0.607√(2/b)
Rayleigh
a
a + √(b/2)
z


p(z)
K
Erlang (Gamma)
K = a(b − 1)^(b−1)/(b − 1)! e^−(b−1)
(b − 1)/a
z


p(z)
a
Exponential
z


p(z)
1/(b − a)
Uniform
a
b
z


p(z)
1 − (P_s + P_p)
P_s
P_p
Salt-and-pepper
0
V
2^k − 1
z

Typical Real-World Sources

Source	Noise Type	Example
Electronic sensor (CCD, CMOS)	Gaussian	Digital cameras
Transmission channel	Salt-and-Pepper	Bit errors
Low photon count	Poisson	X-ray, MRI
Coherent imaging	Speckle	Ultrasound, radar

Statistical Characterization

- Mean (μ): Expected value of noise intensity.

$$E[\eta(x,y)] = \mu$$

- For zero-mean noise, $\mu = 0$.

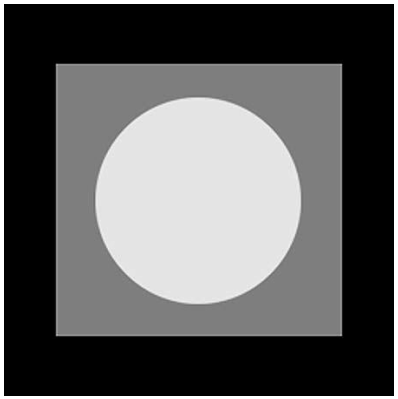
- Variance (σ^2): Spread of noise power.

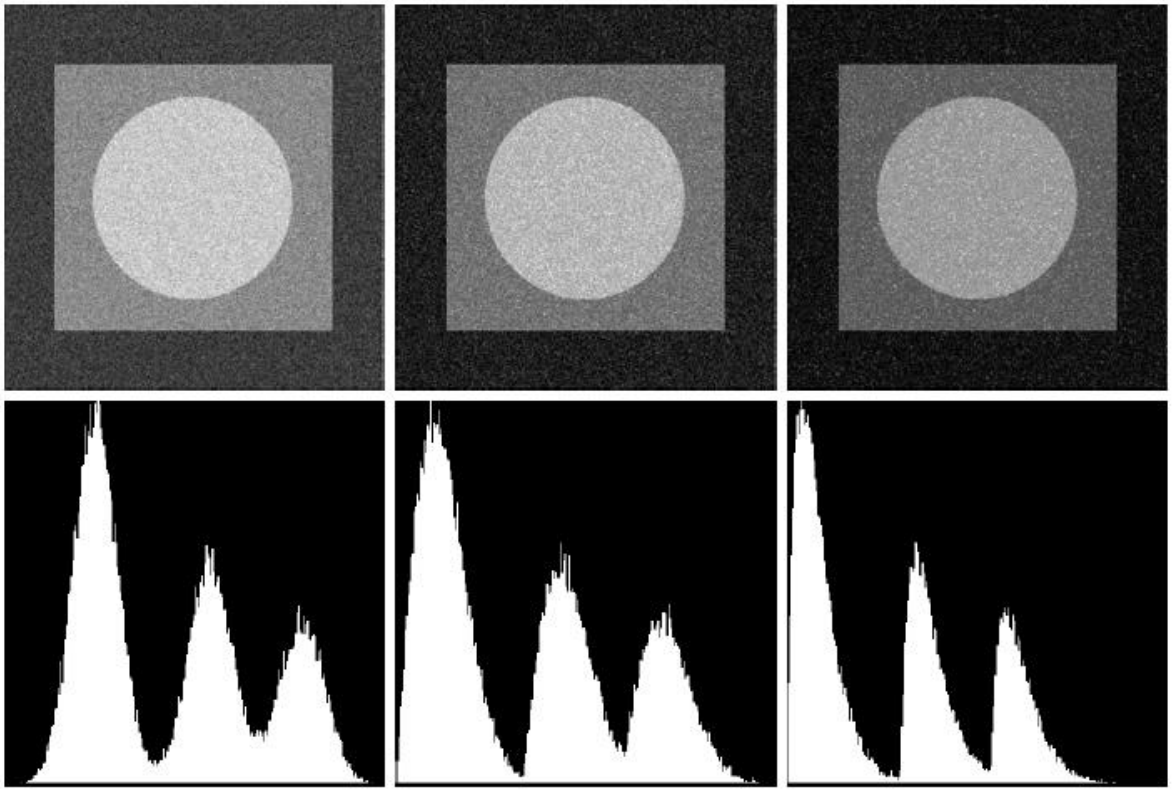
$$Var[\eta(x,y)] = E[(\eta(x,y) - \mu)^2]$$

- Signal-to-Noise Ratio (SNR):

$$SNR = 10\ log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$$

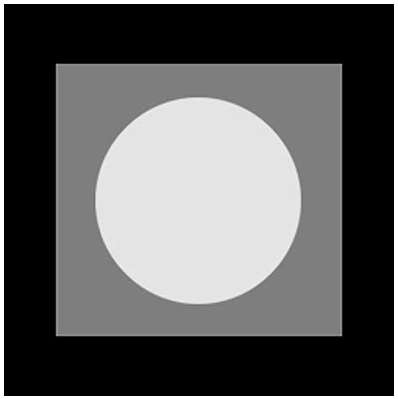
EXAMPLE: Noisy images and their histograms

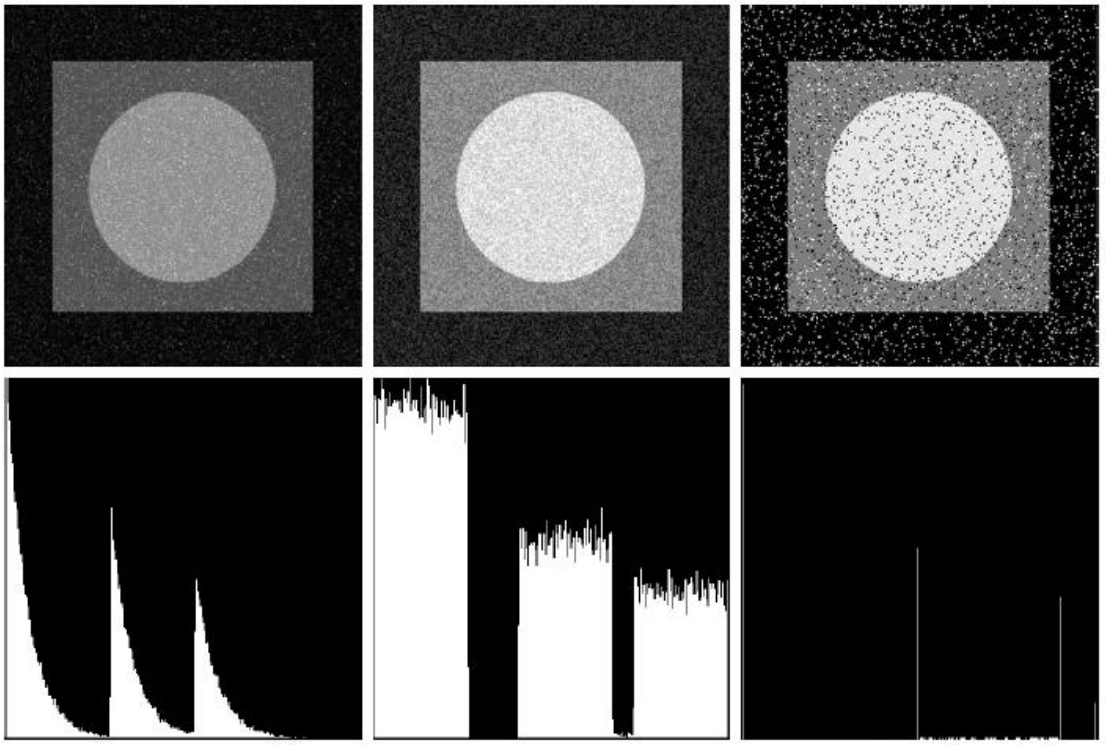




Gaussian Rayleigh Gamma

EXAMPLE: Noisy images and their histograms





Exponential Uniform Salt & Pepper

Removes Salt-and-Pepper Noise by Replacing Pixel with Median of Neighbors

	1	2	3	4	5
1	52	54	53	55	51
2	50	49	255	50	48
3	51	50	49	52	50
4	50	48	47	0	49
5	49	51	50	48	50

Step 1: Select noisy pixel at (2,3) = 255
3×3 neighborhood around (2,3):

54	53	55
49	255	50
50	49	52

Step 2: Sort the neighborhood values
Values: 49,50,49,50,52,53,54,55,255
Sorted: 49,49,50,50,52,53,54,55,255
Median (5th value) = 52

Step 3: Replace pixel (2,3) with median value 52

Step 4: Select noisy pixel at (4,4) = 0
3×3 neighborhood around (4,4):

49	52	50
48	47	0
51	50	48

Removes Salt-and-Pepper Noise by Replacing Pixel with Median of Neighbors

	1	2	3	4	5
1	52	54	53	55	51
2	50	49	255	50	48
3	51	50	49	52	50
4	50	48	47	0	49
5	49	51	50	48	50

Step 5: Sort the neighborhood values

Values: 47,48,49,50,50,51,52,0,48

Sorted: 0,47,48,48,49,50,50,51,52

Median (5th value) = 49

Step 6: Replace pixel (4,4) with median value 49

Position	Original	Median Filtered
(2,3)	255	52
(4,4)	0	49

Conclusion:

- Salt-and-pepper noise pixels (extreme values 0 or 255) get replaced by the median of their neighbors.
- This preserves edges better than averaging (mean filter), as it does not blur edges.

Median filtering is effective for impulse noise removal.

Restoration in Presence of Noise OnlySpatial Filtering

A mean filter smooths local variations in an image, and noise is reduced as a result of blurring.

Arithmetic Mean

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s, t)$$

Restored mean Image

Subimage

Geometric Mean Filter:

multiplication.

$$\hat{f}(x, y) = \left[\prod_{(s,t) \in S_{xy}} g(s, t) \right]^{\frac{1}{mn}}$$

Geometric mean filter achieves smoothing comparable to an arithmetic mean filter, but it tends to lose less image detail in the process.

Restoration in Presence of Noise OnlySpatial Filtering

The harmonic mean filter works well for salt noise, but fails for pepper noise. It does well also with other types of noise like Gaussian noise.

The order of the filter Q is for reducing/virtually eliminating the effects of salt-and-pepper noise. $Q>0$, the filter eliminates pepper noise. $Q<0$, it eliminates salt noise. It cannot do both simultaneously. Note that the contraharmonic filter reduces to the arithmetic mean filter if $Q = 0$, and to the harmonic mean filter if $Q = -1$.

Harmonic Mean Filter:

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s,t)\in S_{xy}} \frac{1}{g(s,t)}}$$

Contraharmonic Mean

$$\hat{f}(x, y) = \frac{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q}}$$

Example: Restoration Techniques

Step 1: Original Image (3×3 pixels)

$$I_{Original} = \begin{bmatrix} 50 & 50 & 50 \\ 50 & 50 & 50 \\ 50 & 50 & 50 \end{bmatrix}$$

All pixels have intensity 50 (uniform).

Step 2: Degraded Image (Add Noise)

Suppose noise affected the image, producing:

$$I_{noisy} = \begin{bmatrix} 48 & 52 & 49 \\ 55 & 200 & 53 \\ 50 & 51 & 48 \end{bmatrix}$$

Notice the **salt noise** at center pixel = 200 (much brighter than others).

Example: Restoration Techniques

$$I_{noisy} = \begin{bmatrix} 48 & 52 & 49 \\ 55 & 200 & 53 \\ 50 & 51 & 48 \end{bmatrix}$$

Step 3: Restoration Using a 3×3 Mean Filter

- Calculate restored value at center pixel (1,1):

$$\frac{1}{9} \times (48 + 52 + 49 + 55 + 200 + 53 + 50 + 51 + 48) = 67.33$$

The restored pixel is closer to original 50, reducing noise effect.

Step 4: Calculate restored pixel at (0,0) with zero padding

Neighbors: assume out-of-bound neighbors are zero.

$$\frac{1}{9} \times (0 + 0 + 0 + 0 + 48 + 52 + 0 + 55 + 200) \approx 39.44$$

Thank You

Lecture 6.2

Image Restoration

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Inverse and Wiener Filtering

Inverse Filtering

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)}$$

Wiener Filtering

$$\hat{F}(u,v) = \frac{H(u,v)}{|H(u,v)|^2 + \frac{S_N(u,v)}{S_F(u,v)}} \times G(u,v)$$

Image Restoration using Inverse Filtering

Suppose you have an original 1D signal:

$$f = [2,3,4]$$

The signal gets blurred by a known **blur kernel** h:

$$h = [1,1]/2 = [0.5,0.5]$$

The blurred observed signal gg is the convolution:

$$g = f * h$$

Calculate g:

$$g0 = 0.5 \times 2 = 1.0 (assuming\ zero\ padding)$$

$$g1 = 0.5 \times 2 + 0.5 \times 3 = 1.0 + 1.5 = 2.5$$

$$g2 = 0.5 \times 3 + 0.5 \times 4 = 1.5 + 2.0 = 3.5$$

$$g3 = 0.5 \times 4 = 2.0$$

So,

$$g = [1.0,2.5,3.5,2.0]$$

Image Restoration using Inverse Filtering

Step 1: Express convolution in matrix form

Since h has length 2, convolution matrix H is:

$$H = \begin{bmatrix} 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 \end{bmatrix}$$

$$f \; is \; a \; 3 \times 1 \; vector, g \; is \; 4 \times 1.$$

Step 2: Inverse filtering to restore *f*

- The Objective: is to find $\hat{f}$ such that

$$g = Hf \implies \hat{f} = H^{-1}g$$

- Since HH is not square, use least squares:

$$\hat{f} = (H^T H)^{-1} H^T g$$

Image Restoration using Inverse Filtering

Step 3: Compute $H^T H$

$$H^T H = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0.25 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0.25 & 0.5 \end{bmatrix}$$

Step 4: Compute $H^T g$

$$H^T g = \begin{bmatrix} 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1.0 \\ 2.5 \\ 3.5 \\ 2.0 \end{bmatrix} = \begin{bmatrix} 1.75 \\ 3.0 \\ 2.75 \end{bmatrix}$$

Image Restoration using Inverse Filtering

Step 5: Solve for $\hat{f} = (H^T H)^{-1} H^T g$

$$(H^T H)^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix}$$

$$\Rightarrow \hat{f} = (H^T H)^{-1} H^T g = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1.75 \\ 3.0 \\ 2.75 \end{bmatrix} = \begin{bmatrix} 2.0 \\ 3.0 \\ 4.0 \end{bmatrix}$$

Step 7: Result

Restored signal:

$$\hat{f} = [2.0, 3.0, 4.0]$$

which matches original *f*.

Minimum MSE using Wiener Filtering

$$G(u) = H(u)F(u) + N(u)$$

- $F(u)$= Fourier transform of the original image (random process)
- $H(u)$= known degradation transfer function (blur)
- $N(u)$= additive noise (zero-mean, independent of F)
- $G(u)$ = observed degraded image spectrum

We seek a linear estimator $\hat{F}(u) = W(u)\,G(u)$ that **minimizes the mean-square error,**

$$MSE = E\{|\,F(u) - \hat{F}(u)\,|^2\}$$

Wiener filter (solution for minimum MSE)

Minimize MSE with respect to complex-valued $W(u)$. The solution (Wiener filter) is:

$$W(u) = \frac{S_F(u)H^*(u)}{|H(u)|^2 S_F(u) + S_N(u)} = W(u) = \frac{H^*(u)}{|H(u)|^2 + \frac{S_N(u)}{S_F(u)}}$$

- $S_F(u) = E\{|\ F(u)\ |^2\}$ is the power spectral density (PSD) of the original image,
- $S_N(u) = E\{|\ N(u)\ |^2\}$ is the PSD of the noise,
- $H^*(u)$ is the complex conjugate of $H(u)$.

This form shows the balance: where $|H(u)|^2$ is large relative to noise-to-signal ratio, the filter approximates the inverse filter $1/H(u)$; where noise dominates, the filter attenuates amplification.

Wiener filter (solution for minimum MSE)

Compute the error spectrum $E(u) = F(u) - \hat{F}(u)$. The error PSD becomes:

$$S_E(u) = E\{|E(u)|^2\}$$

$$= |\,1 - W(u)H(u)\,|^2\, S_F(u) + |\,W(u)\,|^2\, S_N(u)$$

If $W(u)$ is the Wiener filter, the minimum error PSD:

$$S_{E,min}(u) = \frac{S_F(u)\, S_N(u)}{|H(u)|^2\, S_F(u) + S_N(u)}$$

Thus, the total MSE (over all frequencies) is

$$MSE_{min} = \int_{freq} S_{E,min}(u)du$$

(or discrete sum for sampled images).

Interpretation: frequencies where signal power $S_F(u)$ is large or noise $S_N(u)$ small give low error; where noise dominates the error increases.

Wiener filter (solution for minimum MSE)

Algorithm (discrete image, implementable)

1. Compute FFT of degraded image $G[k,\ell]$ and of PSF $H[k,\ell]$ (pad PSF to image size).
2. Estimate PSD $S_F[k,\ell]$ *and* $S_N[k,\ell]$ or assume models (e.g., constant).
3. Build Wiener filter:

$$W[k,\ell] = \frac{S_F[k,\ell]\, H^*[k,\ell]}{|H(k,\ell)|^2 S_F[k,\ell] + S_N[k,\ell]}$$

Or if assumed white and constant $S_F = \sigma_F^2$, $S_N = \sigma_N^2$: use $K = \sigma_N^2/\sigma_F^2$ and

$$W(u) = \frac{H^*}{|H(u)|^2 + K}$$

4. Multiply: $\hat{F} = W \cdot G$.
5. Inverse FFT of $\hat{F} \rightarrow$ clipped/real restored image.

Minimum MSE using Wiener Filtering

Problem Setup

- Original signal $x = [2,4,6,8]$ (unknown)
- Noise $n = [1, -1,0,2]$ (zero mean, variance σ_n^2)
- Observed signal $y = x + n = [3,3,6,10]$

Objective:

Estimate $\hat{x}$ from y using Wiener filtering, minimizing

$$E[(x - \hat{x})^2]$$

Step 1: Assume linear estimator $\hat{x} = wy$ where w is a scalar filter coefficient.

Step 2: Find w minimizing MSE

$$MSE = E[(x - wy)^2]$$

Taking derivative w.r.t. w and setting to zero:

$$\frac{d}{dw} MSE = -2E[xy] + 2wE[y^2] = 0 \implies w = \frac{E[xy]}{E[y^2]}$$

Minimum MSE using Wiener Filtering

$$x = [2,4,6,8], \qquad y = [3,3,6,10]$$

Step 3: Compute expectations from data

- $E[xy] = \frac{1}{4}\sum x_i y_i = \frac{1}{4}(2 \times 3 + 4 \times 3 + 6 \times 6 + 8 \times 10) = 33.5$
- $E[y^2] = \frac{1}{4}\sum x_i y_i = \frac{1}{4}(32 + 32 + 62 + 102) = 38.5$

Step 4: Calculate Wiener filter coefficient

$$w = \frac{33.5}{38.5} \approx 0.8701$$

Step 5: Compute filtered estimate

$$\hat{x} = \boldsymbol{wy} = 0.8701 \times [3, 3, 6, 10]$$
$$= [2.61, 2.61, 5.22, 8.70]$$

Index	Original x	Noisy y	Estimated $\hat{x}$
1	2	3	2.61
2	4	3	2.61
3	6	6	5.22
4	8	10	8.70

The filter smooths noisy observations toward the original signal minimizing MSE.

Thank You

X-ray, CT, and MRI — Image Processing

Feature	X-ray	CT	MRI
Imaging Type/ Energy Source	X-ray radiation 2D image Projection	Tomographic (X-ray based), Tomographic (2D slices, 3D volume)	Tomographic (2D/3D Magnetic Resonance Imaging + Radio waves based)
Data Domain	Spatial, 2D projection intensity	Projection (Radon domain), Multiple projections (0°–180°)	Frequency (k-space), Magnetic resonance signals (frequency domain)
Reconstruction	Direct	Filtered Backprojection	Inverse Fourier Transform
Interaction	Attenuation through tissue	Attenuation from multiple angles	Excitation of hydrogen protons
Best for	Bone, chest, fractures	Bone + soft tissue + 3D	Brain, muscles, soft tissue
Common Processing	Histogram & Edge enhancement	FBP, Wiener, Segmentation	FFT, NLM Denoising, Bias Correction
Output/Key Idea	Shadow image, Measure attenuation X-ray through tissue	Reconstructs cross-sectional images	High-contrast soft-tissue image, Uses nuclear magnetic resonance of hydrogen atoms in tissues

Image Appearance Comparison

- All three create grayscale medical images, but differ in physics, data acquisition, and processing techniques.

Feature	X-ray	CT	MRI
Tissue Contrast	Poor (bones visible)	Moderate (bones + organs)	Excellent (soft tissues, brain, muscles)
Gray Scale	Attenuation only	Density-based (Hounsfield units)	Proton relaxation times (T1, T2)
Resolution	High (2D)	High (3D)	Moderate (depends on sequence)
Scan Time	Seconds	Seconds–Minutes	Several minutes

Image Processing Techniques by Modality

Processing Stage	X-ray	CT	MRI
Noise Reduction	Median / Wiener filter	Wiener, Adaptive, NLM	Gaussian, Anisotropic diffusion, NLM
Enhancement	Histogram Equalization, CLAHE	Edge enhancement, Bone window	Contrast stretching, Bias-field correction
Segmentation	Thresholding, Morphology	Region growing, Watershed, AI segmentation	K-means, Active contours, Deep learning
Reconstruction	Not needed	Filtered Backprojection / Iterative	Inverse Fourier Transform
Artifact Removal	Scatter, Beam Hardening	Motion, Metal artifacts	Field inhomogeneity, Motion correction
Registration & Fusion	Align X-ray pairs	Multi-slice registration	MRI-CT fusion for diagnosis
3D Visualization	Limited	3D volume rendering	3D brain / organ reconstruction

Key Processing Algorithms by Modality

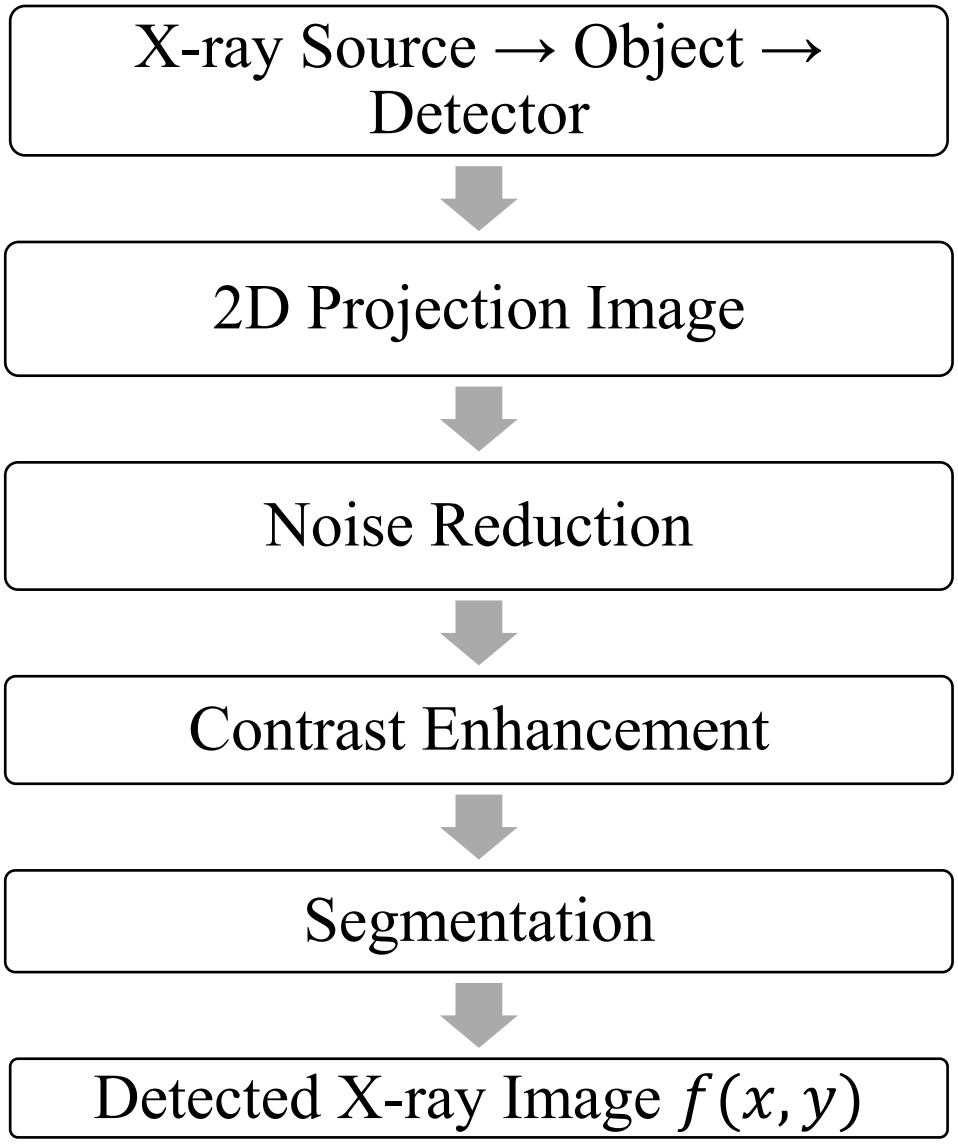
Modality	Common Algorithms
X-ray	Histogram Equalization, CLAHE, Sobel edge, Morphological ops
CT	Radon & Inverse Radon, Ramp filtering, Wiener restoration
MRI	FFT reconstruction, Phase unwrapping, Rician noise reduction, Bias correction

X-ray Image Formation Models

$$f(x, y) = f_0 e^{-\int \mu(x,y,z)dz}$$

- $f(x, y)$: detected intensity,
- f_0: source intensity,
- μ: attenuation coefficient (tissue density).

Image Processing goal: Convert intensity → attenuation map (contrast enhancement, segmentation).



CT Image Computation

CT (Computed Tomography) is an imaging technique that reconstructs a **2D or 3D internal image** of an object (like the human body) from **multiple X-ray projections** taken at different angles.

- CT uses **X-rays** to measure how much radiation is **attenuated** (absorbed) by tissues.
- The computer then **reconstructs** the internal structure from these measurements.

Processing goal: Reconstruct image from projections using **Filtered Backprojection (FBP)** or **Iterative Reconstruction**.

CT Image Computation

X-ray Source → Object → Detector
(rotates around object)

↓

Multiple Projections ($\theta = 0° \rightarrow 180°$)

↓

Sinogram (Radon Transform)

↓

Filtering (Ramp Filter)

↓

Backprojection

↓

Reconstructed CT Image $f(x, y)$

CT Image Formation Model

Let the object's **attenuation coefficient** (how much X-ray is absorbed per unit distance) be $f(x, y)$. Each **X-ray projection** measures how much total attenuation occurs along a straight line through the object. If a ray passes through $f(x, y)$ along a line L, the **projection value** p is:

$$p(\theta, s) = \int_L f(x, y)\, dl$$

$$\Rightarrow p(\theta, s) = \int_{-\infty}^{\infty} f(x, y)\, \delta(r - x\, cos\theta - y\, sin\theta) dx\, dy$$

where

- θ = projection angle (rotation angle of X-ray source)
- s = distance of the ray from the origin

This is called the **Radon Transform** of $f(x, y)$.

CT Image Computation

Radon Transform and Sinogram

When we take many projections (for multiple angles θ):

- Each projection $p(\theta, s)$ is one **line integral** through the object.
- Collect all projections → **sinogram**, which represents all the projection data.

$$p(\theta, s) = R\{f(x, y)\}$$

The Inverse Problem (Reconstruction)

To recover $f(x, y)$ from $p(\theta, s)$, we must perform the **inverse Radon transform** — this is the **CT image reconstruction step**.

CT Image Computation

Two main reconstruction approaches:

A. Analytical Methods: Direct mathematical inversion using **Fourier theory**.

Filtered BackProjection (FBP)/Inverse Radon Transform:

This is the **standard method** in CT.

Steps:

1. Compute 1D Fourier transform of each projection.
2. Multiply by a filter |ω| (ramp filter).
3. Perform inverse Fourier transform → filtered projections.
4. Back projection (sum contributions from all angles).

Mathematically:

$$f(x,y) = \int_0^{\pi} q_\theta\,(r - x\,cos\theta - y\,sin\theta)d\theta$$

$$\Rightarrow f(x,y) = \int_0^{\pi}\int_{-\infty}^{\infty} P(\theta,\omega)|\omega| e^{j2\pi\omega(x\,cos\theta + y\,sin\theta)}\,d\omega\,d\theta$$

where $q_\theta(s)$ is the **filtered projection**.

CT Image Computation

B. Iterative Methods

Numerically refine an estimate of $f(x, y)$:

- Start with an initial guess $f_0(x, y)$
- Compute projections → compare with measured data
- Update image estimate to reduce error

Common methods:

- **ART** (Algebraic Reconstruction Technique)
- **SIRT** (Simultaneous Iterative Reconstruction Technique)
- **MLEM** (Maximum Likelihood Expectation Maximization)

These are slower but handle **noise and incomplete data** better.

CT Image Computation

Fourier Slice Theorem

The **Fourier Slice Theorem** (also called the **Central Slice Theorem**) states:

The **1D Fourier transform** of a **projection** of a 2D input function $f(x, y)$ taken at an angle θ is equal to a **slice** (or line) through the **2D Fourier transform** $F(u, v)$ of that function, taken at the same angle θ in the frequency domain.

Let, $f(x, y)$ be a 2D object (like an image or a cross-section of a body). Its **projection** at angle θ is:

$$P(\theta, r) = \int_{-\infty}^{\infty} f(r\,cos\theta - s\,sin\theta, r\,cos\theta + s\,sin\theta\,)\,ds$$

CT Image Computation

Now take the **1D Fourier Transform** of this projection:

$$P_\theta(\omega) = \int_{-\infty}^{\infty} p_\theta(r) e^{-j2\pi\omega r} dr$$

The **2D Fourier Transform** of the original image is:

$$F(u,v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) e^{-j2\pi(ux+vy)} dx\, dy$$

This theorem connects the projection data to the 2D Fourier Transform of the image:

$$P(\theta,\omega) = F(u,v) = F(\omega\, cos\, \theta\, , \omega \sin\theta)$$

That means, the 1D Fourier transform of the projection $P(\theta,r)$ equals the 2D Fourier transform of $f(x,y)$ evaluated along a radial line (slice) at angle θ. Each 1D projection gives a line in the 2D frequency domain of the image. Once all such lines (for all angles) are filled, applying inverse 2D Fourier transform gives $f(x,y)$.

CT Image Fourier Slice Theorem

$f(x, y)$ *original image*

↓

Multiple Projections

(θ = 0° , 45° , 90° → 180°)

↓

1D Fourier Transform of each projection

↓

Each becomes a radial slice in F(u, v)

↓

Combine slices → Full 2D Fourier Spectrum

↓

Inverse 2D FT → Reconstructed Image

MRI Image Formation Model

- MRI (Magnetic Resonance Imaging) is fundamentally based on nuclear magnetic resonance (NMR) of hydrogen protons in a magnetic field. The goal is to measure spatially encoded signals and reconstruct an image from them. MRI Image Formation Model, from magnetic physics to k-space and finally to image reconstruction using the Fourier Transform.



MRI Image Formation = Magnetic Physics + Fourier Transform Math

Processing goal: Fourier transform reconstruction, artifact correction, denoising, tissue segmentation.

MRI Image Formation Model

- MRI detects the **radio frequency (RF)** signal emitted by hydrogen nuclei (protons) when they return to equilibrium after being excited by an RF pulse inside a **static magnetic field** B_0 (the strong, constant magnetic field at the heart of MRI machines).

Step	Description
Magnetization Alignment	In a strong field B_0, hydrogen nuclei align along or opposite the field.
RF Excitation	A radiofrequency pulse B_1 (weak and time-varying RF field, not a static one) flips the magnetization into the transverse plane.
Signal Emission (Relaxation)	Protons emit RF signals as they relax — this is the MRI signal.
Spatial Encoding	Gradient magnetic fields G_x, G_y, G_z encode spatial position by frequency and phase.
Data Acquisition (k-space)	The signal is sampled in the frequency domain (k-space).
Image Reconstruction	The image is reconstructed via Inverse Fourier Transform (IFT).

MRI Image Formation Model

Mathematical Model

Let $f(x, y)$ represent the **proton density (spin density)** at spatial location (x, y).

The signal received by the MRI coil is:

$$s(t) = \iint f(x, y) e^{-j2\pi(k_x(t)x + k_y(t)y)} dx\, dy$$

Where, $k_x(t) = \int_0^t \gamma G_x(\tau) d\tau$, $k_y(t) = \int_0^t \gamma G_y(\tau)$, and $\gamma =$ gyromagnetic ratio (42.58 MHz/T for Hydrogen).

The **2D Fourier Transform** of $f(x, y)$ is

$$s(k_x, k_y) = \iint f(x, y) e^{-j2\pi(k_x(t)x + k_y(t)y)} dx\, dy$$

Thus, the MRI signal in k-space (frequency domain) directly corresponds to the Fourier transform of the spatial image.

MRI Image Formation Model

Image Reconstruction

The image is reconstructed by taking the **Inverse Fourier Transform**:

$$f(x,y) = \iint s(k_x, k_y) e^{j2\pi(k_x(t)x + k_y(t)y)} dk_x\, dk_y$$

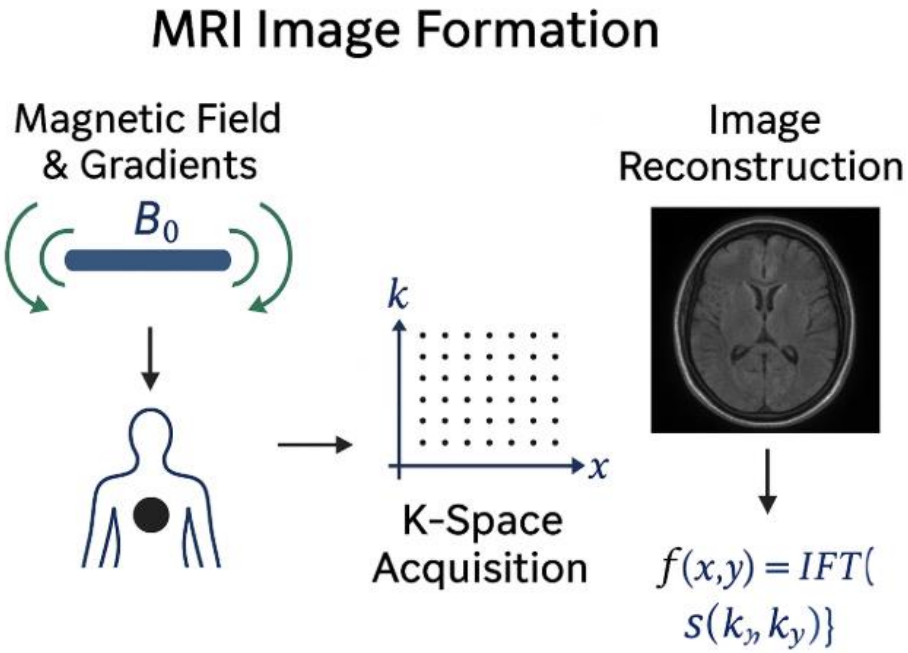
In discrete form (after sampling k-space):

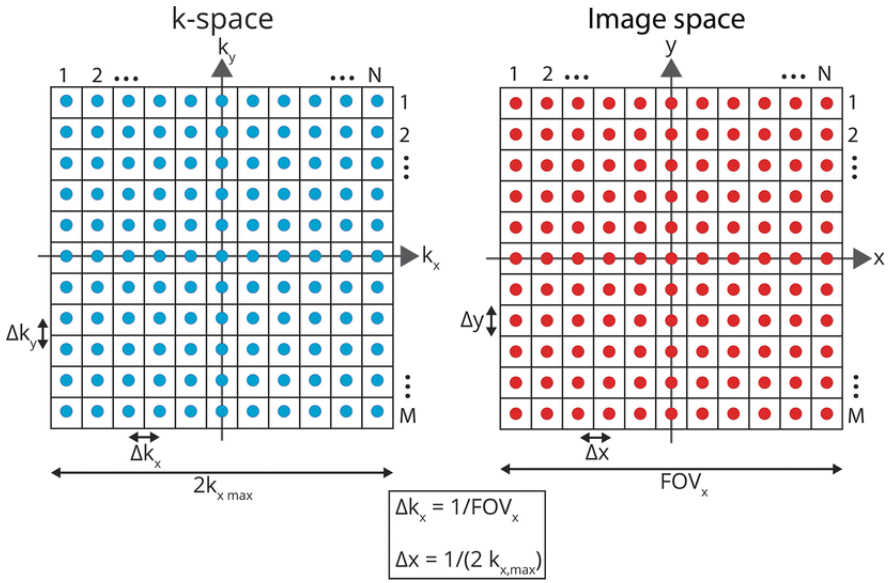
$$f(m,n) = IFFT2\{s[k_x, k_y]\}$$

k-Space Concept

- **k-space** is the **frequency domain** representation of the MRI image.
- Each point in k-space contains information about **the entire image**.
- The **center** of k-space stores **low frequencies** (overall contrast).
- The **edges** of k-space store **high frequencies** (edges, fine detail).

MRI Image Formation Model





MRI Image Formation Model

MRI Signal Components

MRI signals are influenced by:

- Proton density (ρ)
- T1 relaxation time (longitudinal recovery)
- T2 relaxation time (transverse decay)

Thus, the general model is:

$$f(x,y) = \rho(x,y)\left[1 - e^{-\frac{TR}{T_1(x,y)}}\right].e^{-TE/T_2(x,y)}$$

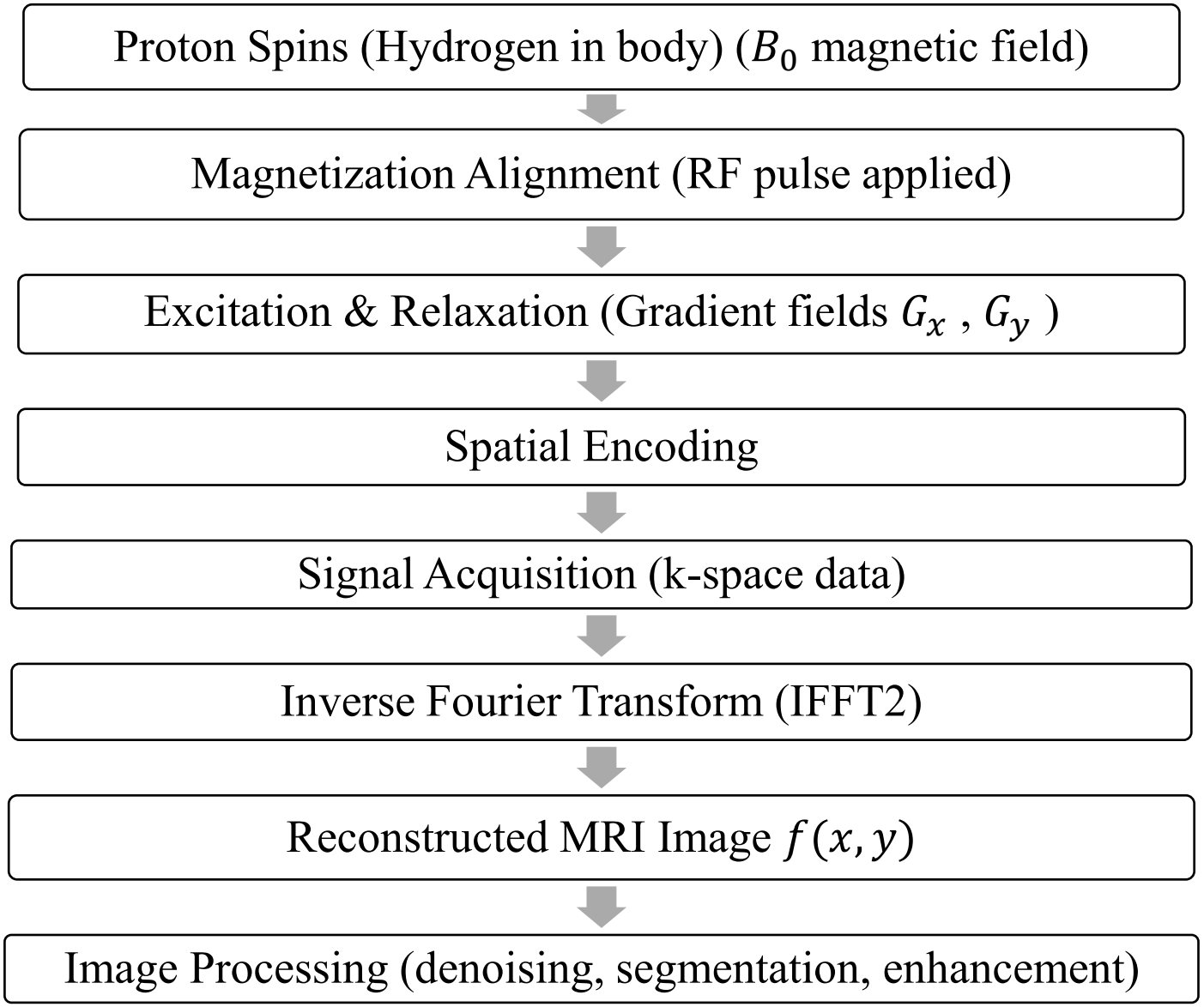
where

- TR: Repetition time
- TE: Echo time

This controls image contrast:

- T1-weighted: short TR, short TE → highlights fat
- T2-weighted: long TR, long TE → highlights fluid
- PD-weighted: long TR, short TE → highlights proton density

MRI Image Formation Model



CT Image Computation

5.6##Consider a very simplified Computed Tomography (CT) image reconstruction example using the Radon transform and filtered back-projection concept on a tiny 2×2 image.

Problem Setup:

Assume the original image $f(x, y)$ is:

$$f = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

We want to reconstruct this image from its projections (sinogram) at two angles: 0° and 90°.

CT Image Computation

Step 1: Compute projections (Radon transform)

•At **0°** (horizontal projection), sum along rows:

Detector position	Sum of pixels
Position 1 (left column)	1 + 3 = 4
Position 2 (right column)	2 + 4 = 6

•At **90°** (vertical projection), sum along columns:

Detector position	Sum of pixels
Position 1 (top row)	1 + 2 = 3
Position 2 (bottom row)	3 + 4 = 7

Step 2: Form sinogram (projections matrix)

Angle (°)	Detector 1	Detector 2
0	4	6
90	3	7

CT Image Computation

Step 3: Back-projection (simple unfiltered)

- For each angle, distribute the projection values back onto the image pixels corresponding to the detector positions.

At 0°:

- Detector 1 (4) adds 4 to pixels in left column → pixels (1,1) & (2,1)
- Detector 2 (6) adds 6 to pixels in right column → pixels (1,2) & (2,2)

At 90°:

- Detector 1 (3) adds 3 to pixels in top row → pixels (1,1) & (1,2)
- Detector 2 (7) adds 7 to pixels in bottom row → pixels (2,1) & (2,2)

CT Image Computation

Step 4: Sum contributions

Pixel	From 0°	From 90°	Total sum
(1,1)	4	3	7
(1,2)	6	3	9
(2,1)	4	7	11
(2,2)	6	7	13

Step 5: Normalize

Because each pixel is counted twice (once per angle), divide by 2:

Pixel	Final estimate
(1,1)	7 / 2 = 3.5
(1,2)	9 / 2 = 4.5
(2,1)	11 / 2 = 5.5
(2,2)	13 / 2 = 6.5

Step 6: Compare with original image

Original	Reconstructed (approx.)
1	3.5
2	4.5
3	5.5
4	6.5

Stroke detection using CT Image

5.7##stroke detection using CT scan image. Detect two common acute-stroke findings on a non-contrast head CT:

(A) acute intracerebral hemorrhage (hyperdense blood), and

(B) early ischemic stroke (subtle hypoattenuation / loss of gray–white differentiation).

A tiny synthetic CT slice (HU values) so every arithmetic step is simple. A cite authoritative radiology sources for the key facts and the CT signs may be used. Note: A real clinical diagnosis must be made by trained radiologists with full images and clinical information.

Solution:

i) **Acute intracerebral hemorrhage** appears **hyperdense** on non-contrast CT (blood often ~50–80+ HU depending on age/hematocrit). Hyperdense focal collections or hyperdense vessels are strong signs of hemorrhage/ thrombus.

ii) **Early ischemic stroke** can cause **subtle hypoattenuation** and **loss of gray–white differentiation** (eg. loss of the insular ribbon, effacement of sulci). These changes are often small (~a few HU) and are best detected by comparing symmetric regions.

Stroke detection using CT Image

	0	1	2	3	4
0	36	35	34	36	35
1	37	75	35	36	34
2	28	29	33	34	35
3	35	36	34	36	35
4	34	35	33	35	34

Here is the 5×5 HU matrix (rows 0…4, cols 0…4):
It can be seen that the hemorrhage at (1,1) = 75 HU, and the hypoattenuation (ischemia) at (2,0),(2,1).

HU: Hounsfield units

- Normal cortical/white matter: ~33–37 HU (we simplified).
- Hemorrhage pixel set to **75 HU** (obviously much higher).
- Ischemic pixels reduced to 28–29 HU (~6–9 HU lower than contralateral).

Stroke detection using CT Image

A. Detecting hemorrhage (hyperdensity)

~>50 HU: strongly suggests acute blood on non-contrast CT.

1. **Scan the matrix for unusually high HU** values. Look at **75 HU** at (1, 1) but all other pixels are in 28–37 HU. **Thus, 75 is an outlier.**

1. **Confirm local neighborhood**: check neighbors of (1,1):
   - ✓ Neighbors: (0,1)=35, (1,0)=37, (1,2)=35, (2,1)=29. All neighbors ~30–37 HU but (1,1) is a focal hyperdensity.

2. **Decision:** This focal hyperdense spot is consistent with **acute intracerebral hemorrhage**. In practice: radiologist would check shape, margins, mass effect, and HU with ROI measurement; if HU >> brain (as here), diagnosis favors blood.

Therefore, the hemorrhage detection is straightforward in this example.

Stroke detection using CT Image

B. Detecting early ischemia (hypoattenuation and loss of gray–white differentiation)

Step 1 — compute mean HU of the suspected region (left peri-insular)
Take the two left pixels at row 2: (2,0)=28 and (2,1)=29.
Mean_left_region = (28 + 29) / 2 = **28.5 HU**.

Step 2 — compute mean HU of the symmetric region on the right
Take the corresponding right-side pixels at row 2: (2,3)=34 and (2,4)=35.
Mean_right_region = (34 + 35) / 2 = **34.5 HU**.

Step 3 — difference and interpretation
Difference = 34.5 − 28.5 = **6.0 HU** (right side is higher **by 6 HU**).

Interpretation: a **~6 HU** focal reduction in attenuation on the left compared with the contralateral side is consistent with early ischemic hypoattenuation.

Stroke detection using CT Image

C. Quick checklist you can do by hand on any small CT matrix

1. **Window / level**: use stroke windows (narrower window) to improve detection of early ischemia. (In real CT software.)

2. **Scan for outliers (hyperdense pixels)** for hemorrhage (>~50–60 HU).

3. **Compute contralateral means** in suspicious regions and take the difference; >~3–5 HU difference in a cortical region + loss of cortical definition is suspicious for acute ischemia.

4. **Look for specific signs:** hyperdense vessel, loss of insular ribbon, sulcal effacement, lentiform nucleus hypoattenuation.

Stroke Detection using CT Image

D. Limitations & clinical cautions (important)

- **Small HU differences are subtle** and easily missed; CT sensitivity is limited in hyperacute ischemia. Early ischemia may be invisible for some time.

- **Mimics exist:** calcified vessels, beam hardening, artifacts can mimic hyperdensity; chronic encephalomalacia or old infarct can mimic hypoattenuation. Always correlate with clinical history and additional imaging (CT angiography, CT perfusion, MRI-DWI).

- This worked example is **simplified** (single-slice, pixelized HU values). Real CT is higher resolution and radiologists use ROIs, windowing, and automated tools (ASPECTS scoring, AI) to help detect early stroke signs.

MRI Image Processing

5.8##Brain Tumor Detection using an MRI image with a small 5×5 grayscale MRI patch to illustrate every step conceptually.

1. MRI Image (Data)

Take a small portion of an MRI slice (grayscale values: 0–255).

- Darker region (40–50): **normal brain tissue**
- Brighter region (160–215): **possible tumor area** (abnormal intensity)

Pixel (x,y)	1	2	3	4	5
1	40	42	45	47	50
2	43	46	160	180	190
3	44	48	170	200	210
4	45	50	175	205	215
5	47	52	60	65	70

MRI Image Processing

2. Preprocessing Step

(a) Noise Reduction

Apply **Median Filter** (3×3 window), at pixel (3,3)

Neighborhood:

$$46, 160, 180$$
$$48, 170, 200$$
$$50, 175, 205$$

Sorted values = [46, 48, 50, 160, 170, 175, 180, 200, 205]

Median = 170

→ Replace center pixel (3,3) with 170.

Result: noise reduced, edges preserved.

MRI Image Processing

3. Intensity Normalization

Normalize image values to 0–1 range:

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

Here, I_{min}=40, I_{max}=215:

- Pixel 40 → (40−40)/(215−40)=0
- Pixel 200 → (200−40)/175≈0.91
- Pixel 215 → 1.0

Thus, tumor area intensity ≈ 0.9–1.0 (bright region).

MRI Image Processing

4. Tumor Region Segmentation

We can apply **Thresholding** to separate bright regions (tumor) from normal tissue.

Threshold value T=0.7.

$$f(x,y) = \begin{cases} 1, & I_{norm}(x,y) > 0.7 \\ 0, & otherwise \end{cases}$$

Let's apply manually:

Pixel	1	2	3	4	5
1	0	0	0	0	0
2	0	0	1	1	1
3	0	0	1	1	1
4	0	0	1	1	1
5	0	0	0	0	0

Tumor pixels (value = 1)

Thus, the **bright cluster in center-right** is detected as tumor region.

MRI Image Processing

5. Morphological Operations

To remove isolated noise points or fill holes:

•Apply **Dilation** then **Erosion** (Morphological Closing)

→ Fills small gaps inside tumor.

Resulting **binary mask** (clean tumor area):

Pixel	1	2	3	4	5
1	0	0	0	0	0
2	0	1	1	1	0
3	0	1	1	1	0
4	0	1	1	1	0
5	0	0	0	0	0

MRI Image Processing

6. Tumor Feature Extraction

Now we compute features from the detected tumor region:

Feature	Formula	result
Area	Number of 1's	9 pixels
Perimeter	Boundary pixels	~8 pixels
Centroid	Mean of (x, y) coordinates of tumor pixels	(3,3.5)
Mean intensity	Average of original values in tumor area (≈190)	190
Shape	Compact cluster → round	—

MRI Image Processing

7. Tumor Classification

If we had multiple MRI samples:

- Extract features (area, intensity, texture)
- Feed into an ML classifier (SVM / CNN)
- Label as **benign** or **malignant**
- But Tumor region is bright, compact → likely **benign**
- If irregular or spread → possibly **malignant**

MRI Image Processing

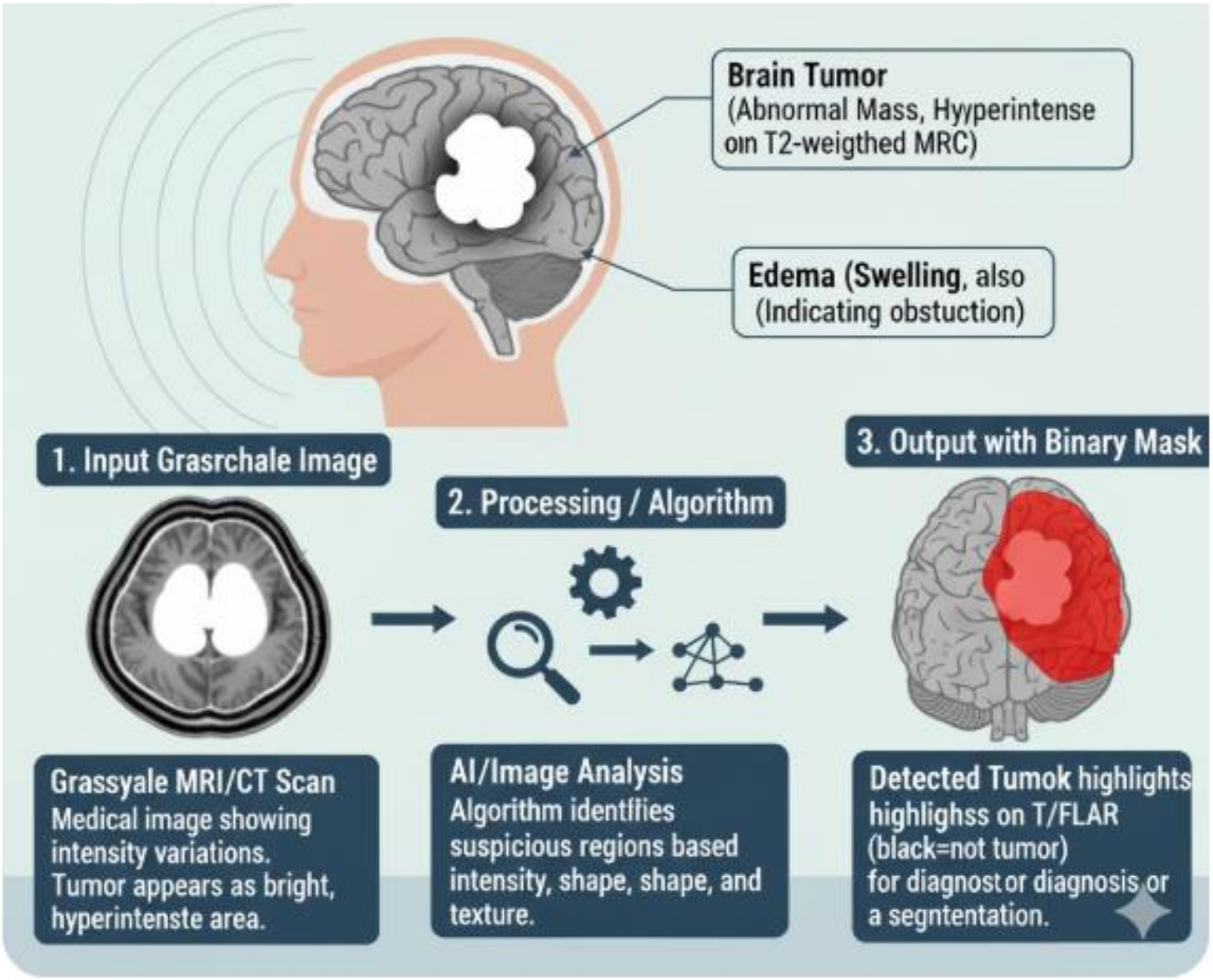
Conceptual Diagram (Text)

MRI Image → Preprocessing → Thresholding
↓
Tumor Segmentation
↓
Morphological Cleaning
↓
Feature Extraction (area, mean, shape)
↓
Tumor Classification (Benign/Malignant)

MRI Image Processing

Step	Operation	Description
1	Input MRI (5×5)	Grayscale brain image
2	Median filter	Noise reduction
3	Normalization	Scale 0–1
4	Thresholding	Segment tumor
5	Morphology	Clean binary mask
6	Feature extraction	Area, centroid, intensity
7	Classification	Identify tumor type

MRI Image Processing



Ultrasound (USG) Image Processing

Ultrasound (USG) image processing is a specialized area of medical image processing. It involves techniques to **enhance, analyze, and interpret ultrasound images** for diagnostic purposes. Ultrasound images are particularly challenging because they are affected by **speckle noise**, low contrast, and artifacts.

2. Preprocessing

a) Speckle Noise Reduction

Since speckle noise is multiplicative, traditional linear filters like mean filters are not ideal. Common methods:

1. **Median Filtering**
   - ✓ Good for removing salt-and-pepper noise but also partially effective for speckle.
   - ✓ Example (3×3 kernel):

     Image patch: Sort values → Median

     [12 15 14] [10 12 14] → 12

     [10 18 12]

     [14 13 15]

Ultrasound (USG) Image Processing

1. **Wiener Filtering** (Adaptive)
   - ✓ Estimates local mean and variance.

   Formula:

$$\hat{I}(x,y) = \mu + \frac{\sigma^2 - v^2}{\sigma^2}(I(x,y) - \mu)$$

where μ and σ^2 are local mean and variance, v^2 is noise variance.

1. **Anisotropic Diffusion**
   - ✓ Preserves edges while reducing noise.
   - ✓ Iterative process:

$$\frac{\partial I}{\partial t} = div(c(x,y,t)\nabla I)$$

Ultrasound (USG) Image Processing

b) Contrast Enhancement

- Ultrasound images often have low contrast.

1. **Histogram Equalization**
   - ✓ Stretch pixel intensities to cover full range.

2. **CLAHE (Contrast Limited Adaptive Histogram Equalization)**
   - ✓ Enhances small regions without amplifying noise excessively.

Ultrasound (USG) Image Processing

3. Segmentation

Segmentation is critical for identifying organs or abnormalities (e.g., tumors, cysts, stones).

1. **Thresholding**
   - ✓ Simple, often used when target structures have different intensity.
   - ✓ Limitation: Ultrasound intensity is often non-uniform.

2. **Edge Detection**
   - ✓ **Sobel**, **Canny** edge detectors.
   - ✓ Can highlight organ boundaries, but sensitive to speckle.

3. **Region-Based Methods**
   - ✓ **Region Growing**: Start from a seed point and grow based on intensity similarity.
   - ✓ **Watershed Segmentation**: Treats intensity as a topographic surface.

4. **Active Contours (Snakes)**
   - ✓ Iteratively deform a curve to fit object boundaries.

Ultrasound (USG) Image Processing

4. Feature Extraction

- Features can be extracted for classification.

1. Texture Features

- ✓ Grey-Level Co-occurrence Matrix (GLCM): Contrast, Correlation, Energy, Homogeneity.

2. Shape Features

- ✓ Area, Perimeter, Compactness.

3. Frequency Domain Features

- ✓ Wavelets or Fourier transforms for speckle patterns.

5. Classification

- Once features are extracted, machine learning or deep learning can classify tissue as normal or abnormal.
- Methods:
  - ✓ SVM, Random Forest, k-NN (classical)
  - ✓ CNNs or U-Net (deep learning for segmentation)

Ultrasound (USG) Image Processing

6. Example Workflow (/ conceptual)

Let's say you have a small 5×5 USG patch:

Original patch (grayscale):

[30 32 33 35 34]
[28 30 31 32 33]
[27 28 29 30 32]
[26 27 28 29 31]
[25 26 27 28 30]

Step 1: Noise Reduction (Median filter 3×3)

- Take median of each 3×3 neighborhood:

Filtered patch:

[30 31 32 33 34]
[28 29 30 31 32]
[27 28 29 30 31]
[26 27 28 29 30]
[25 26 27 28 29]

Step 2: Contrast Enhancement (Simple linear stretch)

- Min=25, Max=34, scale 0–255:

$$I_{new} = \frac{I - 25}{34 - 25} \times 255$$

Step 3: Segmentation (Thresholding)

- Threshold > 150 → target structure

Ultrasound (USG) Image Processing

7. Advanced Techniques

- **Speckle Reduction with Wavelet Transform**
  - ✓ Decompose image → threshold wavelet coefficients → reconstruct.
- **Deep Learning**
  - ✓ U-Net for segmentation, CNNs for feature extraction.
- **3D Ultrasound Processing**
  - ✓ For volumetric organ analysis.

Thank You